

FINITE DIFFERENCE METHODS

Numerical Solution of a Reaction-Diffusion PDE: Design, Analysis, and Results

Analytical and numerical study of a linear reaction-diffusion equation

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Abstract

We study the numerical solution of the linear reaction-diffusion equation

$$u_t = u_{xx} - \alpha u, \quad x \in (0, 1), t > 0,$$

subject to homogeneous Dirichlet boundary conditions and two distinct initial profiles. Three finite-difference schemes are implemented and compared: the explicit Forward-Time Centred-Space (FTCS) method, the implicit Backward-Time Centred-Space (BTCS) method, and the Crank-Nicolson (CN) scheme. Stability is analysed via von Neumann's method; accuracy is verified against the closed-form solution for [f1](#) and the Fourier-series solution for [f2](#). Theoretical consistency gives $O(\Delta t + \Delta x^2)$ for FTCS and BTCS and $O(\Delta t^2 + \Delta x^2)$ for CN. The experiments confirm second-order spatial convergence for all schemes and the expected temporal orders for BTCS and CN, while also illustrating the qualitative difference between the conditionally stable explicit scheme and the unconditionally stable implicit schemes.

1. Problem Formulation

1.1 Governing Equation

The PDE to be solved is

$$u_t(x, t) = u_{xx}(x, t) - \alpha u(x, t), \tag{1}$$

where $\alpha \geq 0$ is a reaction (decay) coefficient. This is a *reaction-diffusion* equation: diffusion spreads spatial gradients while the reaction term $-\alpha u$ damps the amplitude uniformly in space.

Boundary conditions (Dirichlet, homogeneous):

$$u(0, t) = 0, \quad u(1, t) = 0. \tag{2}$$

Initial condition:

$$u(x, 0) = f(x). \tag{3}$$

Two initial profiles are considered:

Table 1. Initial conditions considered in the report.

Label	Definition
f1	$f(x) = \sin(2\pi x)$
f2	$f(x) = 2x$ if $x \leq \frac{1}{2}$, $2(1-x)$ if $x > \frac{1}{2}$

f1 is a pure sine mode and admits a closed-form solution; **f2** is a piecewise-linear tent function whose Fourier decomposition contains infinitely many odd modes.

1.2 Exact Solution for f1

Since $\sin(2\pi x)$ is an eigenfunction of $-\partial_{xx}$ on $(0, 1)$ with eigenvalue $\lambda = 4\pi^2$, substituting $u(x, t) = g(t)\sin(2\pi x)$ into (1) gives

$$g'(t) = -(4\pi^2 + \alpha)g(t) \Rightarrow g(t) = e^{-(4\pi^2 + \alpha)t}. \quad (4)$$

Hence the exact solution is

$$u(x, t) = e^{-(4\pi^2 + \alpha)t} \sin(2\pi x). \quad (5)$$

The solution decays exponentially at rate $4\pi^2 + \alpha \approx 39.5 + \alpha$. Larger α accelerates amplitude decay without altering the spatial shape. This is a key insight: the reaction term does **not** couple spatial modes — it only amplifies the natural exponential decay already present from diffusion.

1.3 Behaviour for f2

Fourier sine series and derivation of coefficients. We expand f_2 in the orthogonal sine basis $\{\sin(n\pi x)\}_{n \geq 1}$ on $(0, 1)$:

$$f_2(x) = \sum_{n=1}^{\infty} b_n \sin(n\pi x), \quad b_n = 2 \int_0^1 f_2(x) \sin(n\pi x) dx. \quad (6)$$

Splitting at $x = 1/2$ and using $f_2(x) = 2x$ on $[0, \frac{1}{2}]$ and $2(1-x)$ on $(\frac{1}{2}, 1]$:

$$b_n = 2 \left[\int_0^{1/2} 2x \sin(n\pi x) dx + \int_{1/2}^1 2(1-x) \sin(n\pi x) dx \right]. \quad (7)$$

Each integral is evaluated by parts, using $\int x \sin(ax) dx = -\frac{x}{a} \cos(ax) + \frac{1}{a^2} \sin(ax)$. For the first piece:

$$\int_0^{1/2} 2x \sin(n\pi x) dx = \left[-\frac{2x}{n\pi} \cos(n\pi x) + \frac{2}{n^2\pi^2} \sin(n\pi x) \right]_0^{1/2} \quad (8)$$

$$= -\frac{1}{n\pi} \cos \frac{n\pi}{2} + \frac{2}{n^2\pi^2} \sin \frac{n\pi}{2}.$$

For the second:

$$\begin{aligned} \int_{1/2}^1 2(1-x) \sin(n\pi x) dx &= \left[-\frac{2(1-x)}{n\pi} \cos(n\pi x) - \frac{2}{n^2\pi^2} \sin(n\pi x) \right]_{1/2}^1 \\ &= \frac{1}{n\pi} \cos \frac{n\pi}{2} + \frac{2}{n^2\pi^2} \sin \frac{n\pi}{2} - \frac{2}{n^2\pi^2} \sin(n\pi). \end{aligned} \quad (9)$$

Summing both pieces and simplifying:

$$b_n = \frac{4}{n^2\pi^2} \left[2\sin \frac{n\pi}{2} - \sin(n\pi) \right] = \frac{8}{n^2\pi^2} \sin \frac{n\pi}{2}. \quad (10)$$

For **even** n : $\sin(n\pi/2) = 0$, so $b_n = 0$.

For **odd** $n = 2k + 1$: $\sin\left(\frac{(2k+1)\pi}{2}\right) = (-1)^k$, so

$$b_{2k+1} = \frac{8}{(2k+1)^2\pi^2} (-1)^k, \quad k = 0, 1, 2, \dots \quad (11)$$

Hence the Fourier sine series is

$$f_2(x) = \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} \sin((2k+1)\pi x). \quad (12)$$

Solution for f2. Each mode $\sin(n\pi x)$ evolves independently; applying the decay rate $n^2\pi^2 + \alpha$ gives

$$u(x, t) = \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} e^{-((2k+1)^2\pi^2 + \alpha)t} \sin((2k+1)\pi x). \quad (13)$$

In practice the series is truncated at $K \approx 50$ terms, since the amplitude of mode $n = 2k + 1$ decays as $(2k + 1)^{-2}$ and the exponential suppresses high modes rapidly for $t > 0$.

Qualitative behaviour. The $k = 0$ (fundamental) mode $\sin(\pi x)$ has the slowest decay rate $\pi^2 + \alpha \approx 9.87 + \alpha$; all higher modes decay at least $9 \times$ faster. By $t \approx 0.1$ the solution is already dominated by $\sin(\pi x)$ regardless of the initial shape. The discontinuity in the first derivative of [f2](#) at $x = 1/2$ slows the decay of Fourier coefficients and makes early-time smoothing more visible on coarse grids, which is a useful test of the schemes' robustness.

2. Grid and Discretization

2.1 Grid

We introduce a uniform spatial step $\Delta x = 1/(N_x + 1)$ and time step $\Delta t = T/N_t$:

$$x_i = i \Delta x, i = 0, \dots, N_x + 1; \quad t^n = n \Delta t, n = 0, \dots, N_t. \quad (14)$$

The N_x interior nodes $x_1, \dots, x_{N_x}$ carry the unknowns; boundary values $u(x_0, t) = u(x_{N_x+1}, t) = 0$ are enforced exactly and are **not stored**.

The *mesh Fourier number* (sometimes called the diffusion number or CFL for diffusion) is

$$r = \frac{\Delta t}{\Delta x^2}. \quad (15)$$

This single dimensionless parameter controls both accuracy and stability of the explicit scheme, and influences the conditioning of the linear systems in the implicit schemes.

2.2 Second-Order Central Difference for u_{xx}

Applying a standard Taylor expansion:

$$u_{xx}(x_i, t^n) \approx \frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{\Delta x^2} + O(\Delta x^2). \quad (16)$$

All three schemes use this same spatial stencil; they differ only in which time level is used to evaluate the right-hand side.

2.3 Discrete Operator, Consistency, and Boundary Treatment

It is useful to define the discrete Laplacian on the vector of interior unknowns:

$$(L_h U)_i = \frac{U_{i-1} - 2U_i + U_{i+1}}{\Delta x^2}, \quad i = 1, \dots, N_x, \quad (17)$$

with $U_0 = U_{N_x+1} = 0$ imposed by the homogeneous Dirichlet boundary conditions. The semi-discrete spatial operator for the PDE is then

$$A_h = L_h - \alpha I. \quad (18)$$

With this notation, the three time discretisations can be written compactly as

$$\text{FTCS:} \quad U^{n+1} = (I + \Delta t A_h) U^n, \quad (19)$$

$$\text{BTCS:} \quad (I - \Delta t A_h) U^{n+1} = U^n, \quad (20)$$

$$\text{CN:} \quad \left(I - \frac{\Delta t}{2} A_h\right) U^{n+1} = \left(I + \frac{\Delta t}{2} A_h\right) U^n. \quad (21)$$

This operator view makes the relationship between the schemes transparent: FTCS uses an explicit Euler step in time, BTCS uses a backward Euler step, and Crank-Nicolson applies the trapezoidal rule to the same semi-discrete ODE system $U'(t) = A_h U(t)$.

The consistency follows from Taylor expansion. For a smooth solution,

$$\frac{u(x_i, t^{n+1}) - u(x_i, t^n)}{\Delta t} = u_t(x_i, t^n) + O(\Delta t), \quad (22)$$

whereas the backward time difference has the same first-order accuracy at t^{n+1} . The Crank-Nicolson average cancels the first-order time defect and gives second-order temporal consistency. Combined with the centred spatial approximation (16), the global formal orders are therefore

$$\text{FTCS, BTCS: } O(\Delta t + \Delta x^2), \quad \text{CN: } O(\Delta t^2 + \Delta x^2). \quad (23)$$

The boundary conditions affect only the first and last interior equations. For example, at $i = 1$ the stencil contains $U_0^n = 0$, so the first row of the matrix has no extra boundary contribution; similarly, at $i = N_x$, $U_{N_x+1}^n = 0$. If the Dirichlet data were non-zero, these same entries would appear as known source terms on the right-hand side. In the present homogeneous case, the tridiagonal matrices shown later already include the boundary conditions exactly.

3. Numerical Schemes

3.1 Explicit Scheme (FTCS)

Replace u_t by a forward difference and u_{xx} by a centred difference at the current time level n :

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{\Delta x^2} - \alpha u_i^n. \quad (24)$$

Solving for u_i^{n+1} :

$$u_i^{n+1} = u_i^n + r(u_{i-1}^n - 2u_i^n + u_{i+1}^n) - \alpha \Delta t u_i^n. \quad (25)$$

This is a **pure update formula** — no linear system needs to be solved. The computational cost per time step is $O(N_x)$ operations.

Matrix form. Defining $\mathbf{u}^n = (u_1^n, \dots, u_{N_x}^n)^\top$, the update reads

$$\mathbf{u}^{n+1} = A_{\text{FTCS}} \mathbf{u}^n, \quad (26)$$

where A_{FTCS} is the $N_x \times N_x$ tridiagonal matrix

$$A_{\text{FTCS}} = (1 - 2r - \alpha\Delta t)I + rT, \quad T = \text{tridiag}(1, 0, 1). \quad (27)$$

Explicitly:

$$A_{\text{FTCS}} = \begin{bmatrix} 1 - 2r - \alpha\Delta t & r & & & \\ r & 1 - 2r - \alpha\Delta t & r & & \\ & \ddots & \ddots & \ddots & \\ & & r & 1 - 2r - \alpha\Delta t & \end{bmatrix}. \quad (28)$$

The homogeneous Dirichlet conditions $u_0 = u_{N_x+1} = 0$ are incorporated exactly: the boundary terms $r \cdot u_0$ and $r \cdot u_{N_x+1}$ vanish and are absent from the first and last rows.

Local truncation error:

$$\tau_{\text{FTCS}} = O(\Delta t) + O(\Delta x^2). \quad (29)$$

The scheme is *first-order in time* and *second-order in space*.

3.2 Implicit Scheme (BTCS)

Replace u_{xx} at time level $n+1$ (future) rather than n :

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{u_{i-1}^{n+1} - 2u_i^{n+1} + u_{i+1}^{n+1}}{\Delta x^2} - \alpha u_i^{n+1}. \quad (30)$$

Rearranging yields a tridiagonal system for the N_x unknowns $\mathbf{u}^{n+1}$:

$$-r u_{i-1}^{n+1} + (1 + 2r + \alpha\Delta t) u_i^{n+1} - r u_{i+1}^{n+1} = u_i^n. \quad (31)$$

In matrix form $A\mathbf{u}^{n+1} = \mathbf{u}^n$ where

$$A = \text{tridiag}(-r, 1 + 2r + \alpha\Delta t, -r). \quad (32)$$

The matrix A is strictly diagonally dominant (diagonal entry $1 + 2r + \alpha\Delta t > 2r$), hence nonsingular; this also makes the Thomas solve stable without pivoting in the one-dimensional setting considered here.

The **Thomas algorithm** (a specialised Gaussian elimination for tridiagonal systems) solves (32) in $O(N_x)$ operations without pivoting, exploiting the sparsity structure.

Local truncation error: $O(\Delta t) + O(\Delta x^2)$ — same order as FTCS.

3.3 Crank-Nicolson Scheme

CN averages the spatial operator at levels n and $n + 1$:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{1}{2} \left[\frac{\delta_x^2 u_i^n}{\Delta x^2} + \frac{\delta_x^2 u_i^{n+1}}{\Delta x^2} \right] - \frac{\alpha}{2} (u_i^n + u_i^{n+1}), \quad (33)$$

where $\delta_x^2 u_i = u_{i-1} - 2u_i + u_{i+1}$. Rearranging:

Left side (unknowns at $n + 1$):

$$\mathcal{L}_i^{n+1} = -\frac{r}{2} u_{i-1}^{n+1} + \left(1 + r + \frac{\alpha \Delta t}{2}\right) u_i^{n+1} - \frac{r}{2} u_{i+1}^{n+1}. \quad (34)$$

Right side (known at n):

$$\mathcal{R}_i^n = \frac{r}{2} u_{i-1}^n + \left(1 - r - \frac{\alpha \Delta t}{2}\right) u_i^n + \frac{r}{2} u_{i+1}^n. \quad (35)$$

The equality $\mathcal{L}_i^{n+1} = \mathcal{R}_i^n$ gives the matrix form $A_{\text{CN}} \mathbf{u}^{n+1} = B_{\text{CN}} \mathbf{u}^n$, where both A_{CN} and B_{CN} are tridiagonal. More explicitly,

$$A_{\text{CN}} = \text{tridiag}\left(-\frac{r}{2}, 1 + r + \frac{\alpha \Delta t}{2}, -\frac{r}{2}\right), \quad (36)$$

$$B_{\text{CN}} = \text{tridiag}\left(\frac{r}{2}, 1 - r - \frac{\alpha \Delta t}{2}, \frac{r}{2}\right). \quad (37)$$

The right-hand side is evaluated explicitly, then the same Thomas algorithm solves the left-hand tridiagonal system.

Local truncation error: $O(\Delta t^2) + O(\Delta x^2)$.

CN is **second-order in both time and space** — one order higher in time than FTCS/BTCS, at the cost of evaluating both sides of the system per step.

4. Stability Analysis (Von Neumann)

A necessary condition for stability of a linear scheme on a periodic domain is that the amplification factor G satisfies $|G| \leq 1$ for all wavenumbers. We write $u_i^n = G^n e^{i\xi x_i}$ with wavenumber ξ and substitute into each scheme.

4.1 FTCS

Substituting into (25):

$$G = 1 - 4r \sin^2\left(\frac{\xi \Delta x}{2}\right) - \alpha \Delta t. \quad (38)$$

Let $\theta \in [0, \pi]$ with $\theta = \xi \Delta x$. The real amplification factor is

$$G(\theta) = 1 - 4r \sin^2(\theta/2) - \alpha \Delta t. \quad (39)$$

For stability $-1 \leq G \leq 1$:

- Upper bound $G \leq 1$: automatic since $4r \sin^2 \geq 0$ and $\alpha \Delta t \geq 0$.
- Lower bound $G \geq -1$: worst case at $\sin^2(\theta/2) = 1$:

$$1 - 4r - \alpha \Delta t \geq -1 \Rightarrow r \leq \frac{1}{2} - \frac{\alpha \Delta t}{4}. \quad (40)$$

Equivalently, the exact FTCS stability condition for this equation is

$$4r + \alpha \Delta t \leq 2 \iff r \leq \frac{1}{2} - \frac{\alpha \Delta t}{4}. \quad (41)$$

When $\alpha \Delta t$ is very small this is often approximated by the classical heat equation bound $r \leq 1/2$, but for $\alpha > 0$ the exact admissible value of r is slightly smaller. Violating (41) causes the high-frequency mode $\theta = \pi$ to be amplified without bound, producing the overflow behaviour observed numerically (see §6.2).

4.2 BTCS

Substituting into (31):

$$G = \frac{1}{1 + 4r \sin^2(\theta/2) + \alpha \Delta t}. \quad (42)$$

Since the denominator is always ≥ 1 , we have $0 < G \leq 1$ for all $r > 0$, $\alpha \geq 0$. BTCS is **unconditionally stable**.

4.3 Crank-Nicolson

$$G = \frac{1 - 2r \sin^2(\theta/2) - \alpha \Delta t/2}{1 + 2r \sin^2(\theta/2) + \alpha \Delta t/2}. \quad (43)$$

Both numerator and denominator are real. Write $\mu = 2r \sin^2(\theta/2) + \alpha \Delta t/2 \geq 0$. Then $G = (1 - \mu)/(1 + \mu)$, which satisfies $-1 < G \leq 1$ for all $\mu \geq 0$. CN is **unconditionally stable**.

However, for large Δt the factor μ can exceed 1, giving $G < 0$. A negative amplification factor reverses sign at each step, producing temporal oscillations on a $2\Delta t$ period. These oscillations are *stable* (they don't grow) but can look like numerical noise if the time step is much too large. In practice, CN should be used with moderate r to suppress these oscillations.

4.4 Summary

Table 2. Accuracy and stability summary for the three schemes.

Scheme	Order (Δt)	Order (Δx)	Stability
FTCS	1	2	Conditional: $4r + \alpha\Delta t \leq 2$
BTCS	1	2	Unconditional
CN	2	2	Unconditional

5. Implementation

5.1 File Structure

```

main.py
benchmark_optimizations.py # reproducible optimisation benchmark
schemes/
  explicit.py      # FTCS step function
  implicit.py      # Thomas algorithm + BTCS solver
  crank_nicolson.py # CN RHS builder + solver
utils/
  grid.py          # dx, dt, r computation
  initial_conditions.py # f1, f2, exact solution
  solver.py        # dispatch layer
  visualization.py # snapshot plots, validation figures

```

5.2 Thomas Algorithm

For a tridiagonal system with lower diagonal a , main diagonal d , upper diagonal c , and right-hand side b , the Thomas algorithm proceeds as follows.

Forward sweep (eliminate sub-diagonal):

$$c'_0 = c_0/d_0, \quad b'_0 = b_0/d_0, \quad (44)$$

$$c'_i = \frac{c_i}{d_i - a_i c'_{i-1}}, \quad b'_i = \frac{b_i - a_i b'_{i-1}}{d_i - a_i c'_{i-1}}, \quad i = 1, \dots, N_x - 1. \quad (45)$$

Back substitution:

$$x_{N_x-1} = b'_{N_x-1}, \quad x_i = b'_i - c'_i x_{i+1}, \quad i = N_x - 2, \dots, 0. \quad (46)$$

The algorithm requires $2 \times (N_x - 1)$ divisions and $4 \times (N_x - 1)$ multiplications/additions — strictly $O(N_x)$ — and stores only three extra arrays of length N_x . No pivoting is needed because the matrix A (BTCS) and A_{CN} are both strictly diagonally dominant throughout the computation.

For BTCS, $d_i = 1 + 2r + \alpha\Delta t$ and $a_i = c_i = -r$ are constant across all rows, so the modified coefficients c'_i need only be computed once (they are the same for every time step). The implementation pre-computes these pivots and then applies only the right-hand-side sweep at each time step.

5.3 Boundary Condition Enforcement

Interior unknowns $u_1, \dots, u_{N_x}$ are stored; boundary values $u_0 = u_{N_x+1} = 0$ are applied implicitly by setting $a_1 = 0$ and $c_{N_x} = 0$ (i.e., the tridiagonal system already incorporates them in its first and last equations). The explicit update uses guard conditions `left=0` for $i = 0$ and `right=0` for $i = N_x - 1$, identical in effect.

6. Numerical Results

All experiments use $\alpha = 1.0$ unless noted. The plotted "exact" curve is (5) for `f1` and the truncated Fourier series (13) with 50 odd modes for `f2`.

6.1 Stable Comparison for `f1` and `f2`

The main comparison figures use the same stable grid for both initial conditions: $N_x = 50$, $N_t = 3000$, $T = 0.5$, hence $\Delta x = 1/51$, $\Delta t = 1/6000$, and $r = 0.4335$. The exact FTCS stability limit is $0.5 - \alpha\Delta t/4 \approx 0.49996$, so all three schemes are stable in these two figures.

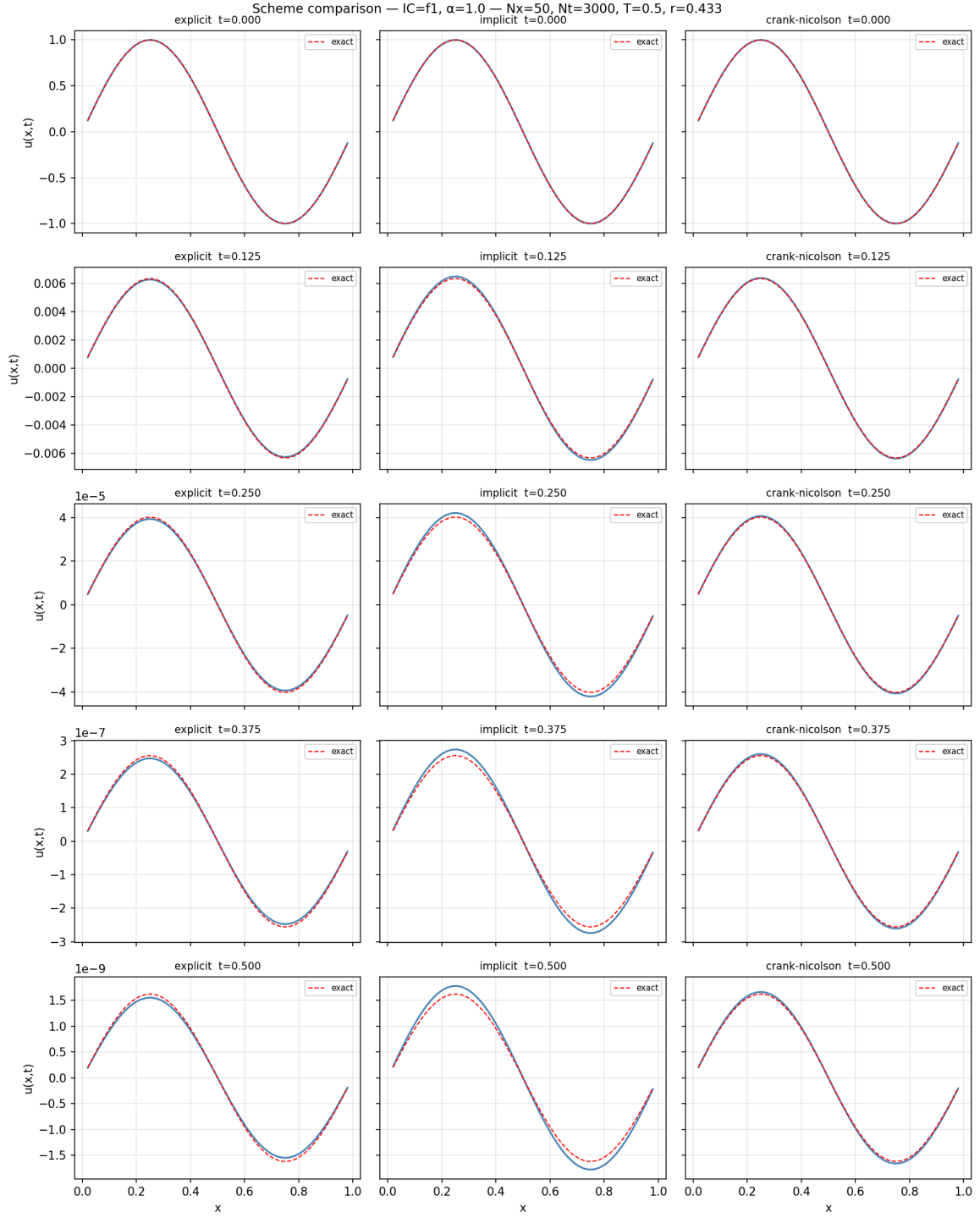


Figure 1. Stable comparison of explicit, implicit, and Crank-Nicolson schemes for f_1 with $N_x=50$, $N_t=3000$, $T=0.5$, $\alpha=1.0$.

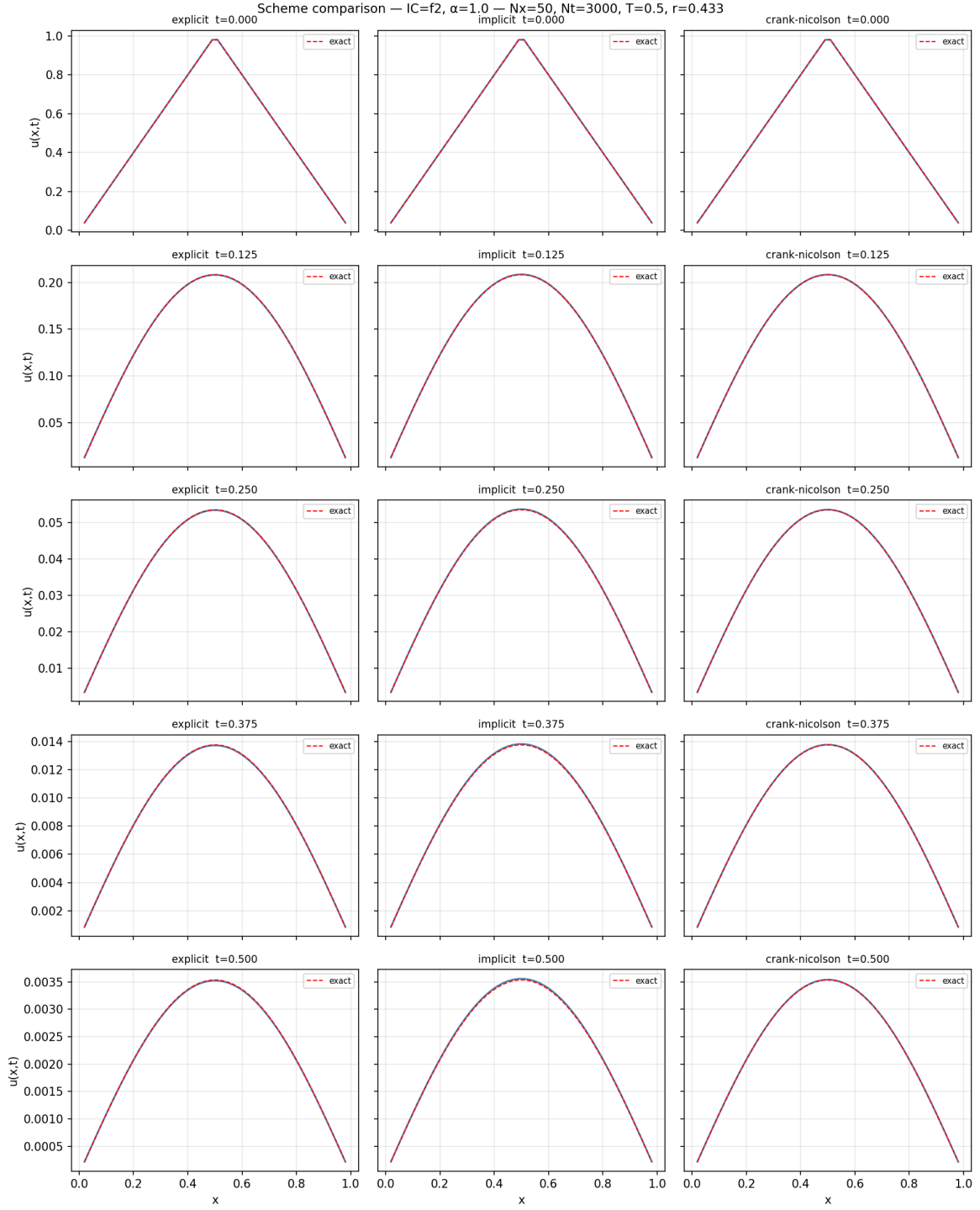


Figure 2. Stable comparison of explicit, implicit, and Crank-Nicolson schemes for f_2 with $N_x=50$, $N_t=3000$, $T=0.5$, $\alpha=1.0$.

For f_1 , the sine shape is preserved while the amplitude decays at the rate $4\pi^2 + \alpha$. For f_2 , diffusion rapidly smooths the tent profile and the solution becomes dominated by the fundamental $\sin(\pi x)$ mode. In both cases the numerical curves are visually indistinguishable from the exact reference at this resolution.

6.2 Validation and Instability Demonstration

The Crank-Nicolson validation figure for `f1` confirms the expected agreement with the exact solution even on the coarser grid $N_x = 100$, $N_t = 500$.

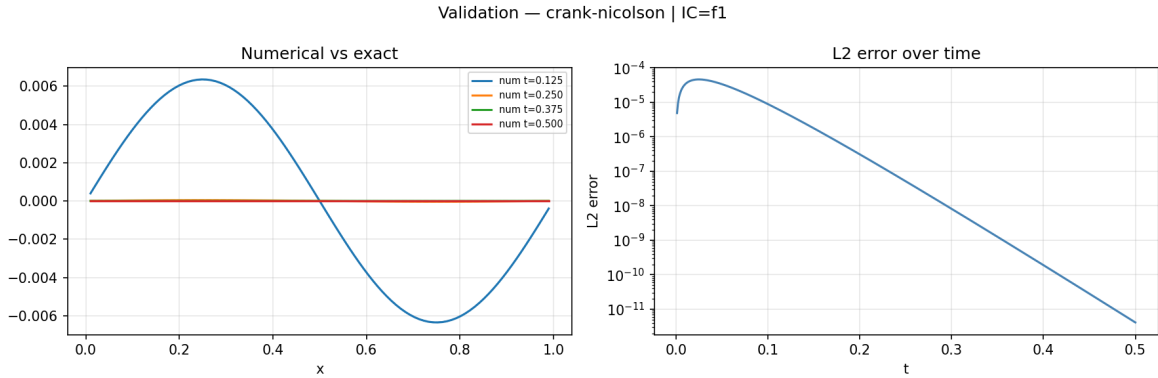


Figure 3. Crank-Nicolson validation against the exact solution for `f1` with $N_x=100$, $N_t=500$, $T=0.5$, $\alpha=1.0$.

For the same coarse time grid, the explicit scheme is deliberately unstable: $N_x = 100$, $N_t = 500$ gives $r = 10.201$, while the exact FTCS limit is about 0.49975 . FTCS therefore overflows, whereas BTCS and CN remain bounded.

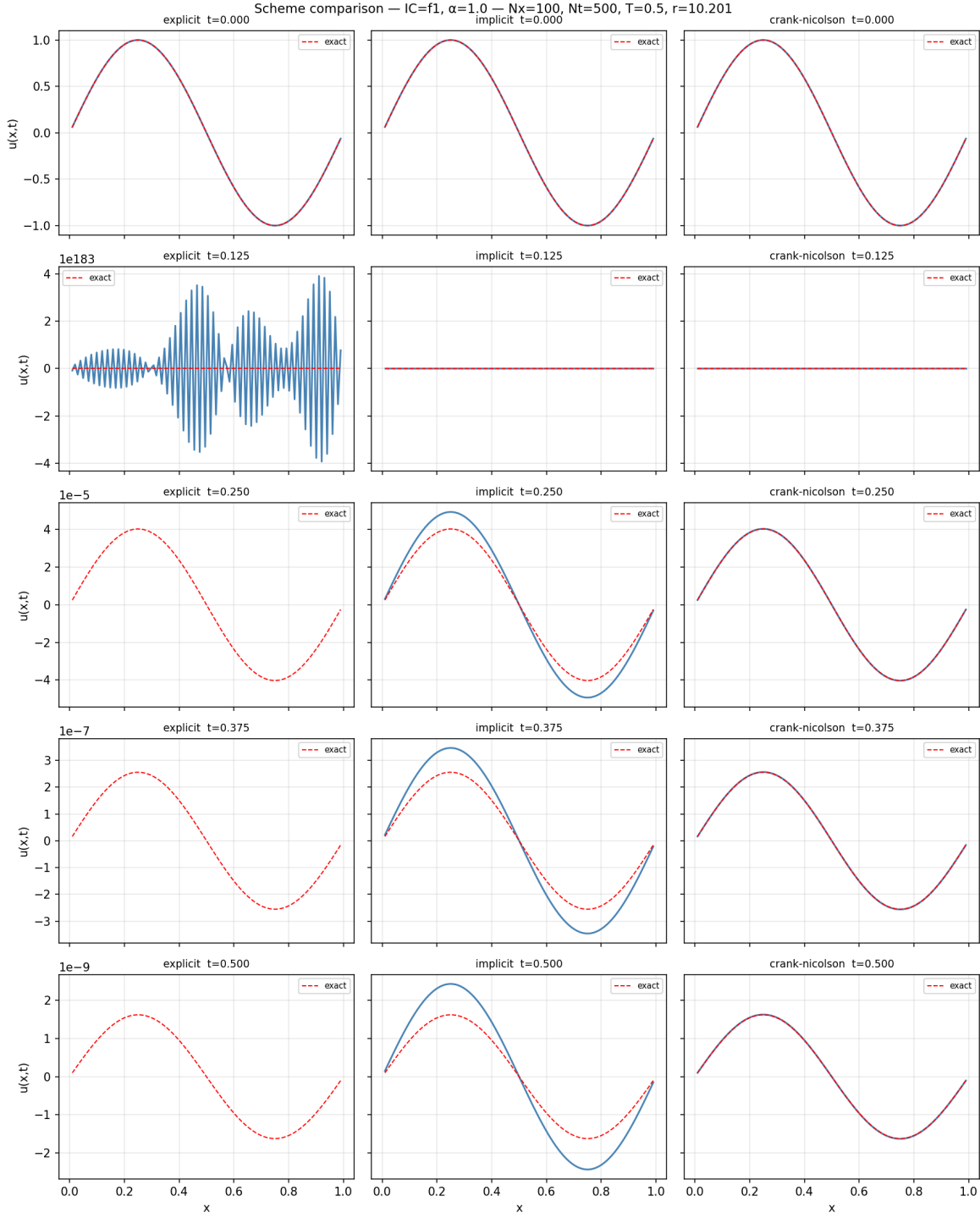


Figure 4. Instability demonstration for f_1 with $N_x=100$, $N_t=500$, $T=0.5$, $\alpha=1.0$. FTCS diverges; BTCS and CN remain stable.

To obtain a stable FTCS run with $N_x = 100$, $T = 0.5$, $\alpha = 1$, the exact bound $4r + \alpha\Delta t \leq 2$ gives

$$N_t \geq \left\lceil 2T(N_x + 1)^2 + \frac{\alpha T}{2} \right\rceil = 10202. \quad (47)$$

This reflects the well-known cost of explicit schemes for diffusion problems: the time step must scale as $\Delta t \sim \Delta x^2$, so halving the spatial step requires roughly four times as many time steps.

6.3 Spatial Convergence (All Schemes)

With $T = 0.1$ and $N_t = 10,000$ (time error negligible), varying N_x shows $O(\Delta x^2)$ convergence for all three schemes, as expected from the second-order central difference:

Table 3. Spatial convergence for the three schemes.

N_x	Δx	FTCS L2	BTCS L2	CN L2	Observed rate
5	0.16667	5.1e-3	5.1e-3	5.1e-3	—
10	0.09091	1.4e-3	1.4e-3	1.4e-3	≈ 1.90
20	0.04762	3.6e-4	3.8e-4	3.7e-4	≈ 1.93
40	0.02439	8.6e-5	1.1e-4	9.6e-5	≈ 1.99
80	0.01235	1.4e-5	3.5e-5	2.4e-5	≈ 2.00

At coarser grids all schemes yield nearly identical errors, confirming that for N_t large enough the spatial discretisation dominates and the schemes are indistinguishable in accuracy.

6.4 Temporal Convergence

With $T = 0.1$ and $N_x = 500$ (spatial error $\lesssim 10^{-6}$, negligible), varying N_t :

Crank-Nicolson (expected $O(\Delta t^2)$):

Table 4. Temporal convergence of Crank-Nicolson.

N_t	Δt	L2 error	Observed rate
5	0.0200	2.69e-3	—
10	0.0100	6.80e-4	1.98
20	0.0050	1.70e-4	2.00
40	0.0025	4.20e-5	2.02
80	0.0013	1.00e-5	2.07

The clean rate of **2.0** confirms second-order accuracy in time for CN.

BTCS (expected $O(\Delta t)$):

Table 5. Temporal convergence of BTCS.

N_t	Δt	L2 error	Observed rate
5	0.0200	2.41e-2	—
10	0.0100	1.13e-2	1.10
20	0.0050	5.38e-3	1.07
40	0.0025	2.61e-3	1.04
80	0.0013	1.29e-3	1.02
160	0.0006	6.38e-4	1.01

The rate converges cleanly to **1.0**.

The temporal error of BTCS at $N_t = 5$ is about $35\times$ larger than CN at the same N_t . To match CN's accuracy at $N_t = 10$ ($L_2 \approx 6.8 \times 10^{-4}$), BTCS needs $N_t \approx 35$, i.e., $3.5 \times$ more steps.

6.5 Scheme Comparison at $N_x = 100$, $N_t = 5,000$

Here $r = 1.02$, which is still above the explicit stability limit. Therefore FTCS is intentionally reported as unstable, while the two implicit schemes remain valid.

Table 6. Final-time errors for f_1 at $N_x = 100$, $N_t = 5000$.

Scheme	L2 error	L^∞ error	Comment
FTCS	NaN	NaN	Unstable ($4r + \alpha\Delta t > 2$)
BTCS	5.5e-11	7.8e-11	First-order in t
CN	7.3e-12	1.0e-11	Second-order in t

The absolute errors are tiny because the exact [f1](#) solution has already decayed to amplitude $\approx 1.6 \times 10^{-9}$ by $T = 0.5$. CN is still about seven times more accurate than BTCS at this grid, consistent with its second-order time accuracy.

6.6 Effect of α

As α increases, the solution decays faster but the spatial structure is unchanged (for [f1](#)). The numerical schemes remain accurate across the tested range:

Table 7. Effect of the reaction coefficient on CN absolute error.

α	$\max \text{abs}(u_{\text{num}})$ at $T = 0.5$	CN L2 error
0.0	2.7e-9	7.2e-12
0.5	2.1e-9	5.5e-12
1.0	1.6e-9	4.1e-12
5.0	2.2e-10	4.2e-13
10.0	1.8e-11	1.7e-14

Larger α damps the solution to near-zero faster, so absolute errors also decrease. This table should therefore be interpreted as an absolute-error check; the explicit stability bound must still include the $\alpha\Delta t$ term.

7. Discussion

7.1 Stability vs. Computational Cost

The FTCS constraint $4r + \alpha\Delta t \leq 2$ forces $\Delta t \leq 2/(4/\Delta x^2 + \alpha)$, which is essentially $\Delta t \lesssim \Delta x^2/2$ when $\alpha\Delta t$ is small. For a spatial resolution of $N_x = 100$ ($\Delta x \approx 0.01$), a stable FTCS simulation of $T = 0.5$ and $\alpha = 1$ requires at least $N_t = 10202$ steps. BTCS and CN need only as many steps as dictated by *accuracy*, not *stability* — and for smooth solutions, accuracy is achieved with far fewer steps.

This disparity grows as N_x increases: doubling the spatial resolution quadruples the required number of FTCS steps (due to the $\Delta t \sim \Delta x^2$ constraint) while BTCS/CN may need only twice as many steps to maintain $O(\Delta t)$ accuracy. For three-dimensional problems, this advantage is even more pronounced.

7.2 Accuracy Trade-Off Between BTCS and CN

Both BTCS and CN solve one tridiagonal system per step. The extra computation in CN is building the right-hand side $B_{\text{CN}}\mathbf{u}^n$, which requires one additional $O(N_x)$ vector pass. Against this modest overhead, CN gains a full order in time accuracy.

The break-even point: to achieve a target temporal error ε in time T , BTCS requires $N_t \sim T\varepsilon^{-1}$ steps while CN needs $N_t \sim \sqrt{T\varepsilon^{-1}}$ steps. For $\varepsilon = 10^{-4}$, $T = 0.5$: BTCS needs $N_t \sim 5000$, CN needs $N_t \sim 71$. The savings factor is ~ 70 , far outweighing the modest extra right-hand-side assembly cost. **CN is the preferred scheme for smooth solutions.**

7.3 Numerical Diffusion

Numerical diffusion (also called numerical dissipation) refers to the extra amplitude damping introduced by the discretisation beyond that predicted by the exact PDE. It can be quantified by comparing the numerical amplification factor G with the exact one $G_{\text{exact}} = e^{-(\lambda + \alpha)\Delta t}$, where $\lambda = 4r\sin^2(\theta/2)/\Delta t$ is the spatial eigenvalue contribution.

BTCS. The backward-Euler time discretisation replaces the exact factor $e^{-(\lambda + \alpha)\Delta t}$ by $1/(1 + (\lambda + \alpha)\Delta t)$. For small $\xi = (\lambda + \alpha)\Delta t$ a Taylor expansion gives

$$\frac{1}{1+\xi} = e^{-\xi} \cdot e^{-\xi^2/2 + O(\xi^3)} \approx e^{-\xi} e^{-\xi^2/2}, \quad (48)$$

meaning BTCS *over-damps* modes by an extra factor $e^{-\xi^2/2}$: each Fourier mode decays faster than the exact solution. This is **first-order numerical diffusion** — the error in the amplification factor is $O(\Delta t^2)$ per step, accumulating to $O(\Delta t)$ over a fixed time T .

FTCS. The forward-Euler factor $1 - \xi = e^{-\xi} e^{\xi^2/2 + O(\xi^3)}$ is the mirror image: modes are *under-damped* when ξ is small and the scheme is still stable. Paradoxically, FTCS introduces *less* numerical diffusion than BTCS near the stability limit but becomes unstable when $4r + \alpha\Delta t > 2$ (approximately $r > 1/2$ when $\alpha\Delta t$ is small).

Crank-Nicolson. The bilinear (Padé) approximation gives

$$\frac{1 - \xi/2}{1 + \xi/2} = e^{-\xi} \cdot e^{\xi^3/12 + O(\xi^5)}, \quad (49)$$

so the extra damping is only $O(\Delta t^3)$ per step, accumulating to $O(\Delta t^2)$ — consistent with second-order accuracy in time. CN introduces negligible numerical diffusion for moderate Δt .

Practical consequence. For the tent function [f2](#), whose high-frequency Fourier modes carry the kink at $x = 1/2$, the excess damping in BTCS causes the kink to smooth out faster than the exact solution. CN preserves the amplitude of each mode much more faithfully. This is directly visible in the comparison plots: BTCS solutions appear slightly smoother than CN solutions at the same N_t .

7.4 When to Use Each Scheme

- **FTCS:** Instructive for learning; useful when $r \ll 1/2$ (very coarse time steps are needed anyway for accuracy, and no linear solve is needed). Can be competitive for nonlinear problems where implicit solves are expensive.
- **BTCS:** Robust and simple; preferred when stability is paramount and temporal accuracy requirements are modest, or as an inner solver in splitting methods.

- **CN**: Best overall accuracy-efficiency for smooth, well-resolved problems. Standard choice in production heat-equation solvers. Should be used with moderate r to avoid temporal oscillations (which are stable but aesthetically undesirable).

7.5 The Reaction Term

A noteworthy observation: the $-\alpha u$ term does not alter the spatial modes, it only adjusts their temporal decay rate from λ_k to $\lambda_k + \alpha$. For numerical purposes, $\alpha > 0$ has two different effects:

- In FTCS, the amplification factor $G = 1 - 4r\sin^2(\theta/2) - \alpha\Delta t$ shifts downward, so the exact stability condition becomes $4r + \alpha\Delta t \leq 2$. Thus the reaction term slightly **tightens** the explicit time-step restriction.
- In BTCS/CN, larger α increases the diagonal dominance of A , improving the conditioning of the tridiagonal system.

This behaviour generalises: for parabolic PDEs with positive reaction, implicit schemes tend to become more strongly damped and often show smaller absolute errors as the reaction coefficient grows. This is the opposite of the difficulty introduced by stiff reaction terms with $\alpha < 0$ (growth) or in systems of PDEs.

7.6 Behaviour with f2 (Tent Function)

The tent function has a discontinuous first derivative at $x = 1/2$, whose Fourier series converges only as $O(k^{-2})$ (rather than the exponential convergence for smooth data). This means:

1. Coarse grids will misrepresent the initial kink.
2. The early-time solution shows the kink smoothing by diffusion — a physically meaningful phenomenon (the PDE regularises the data immediately for $t > 0$).
3. All three schemes handle this correctly, but CN at large Δt can produce visible oscillations near the kink due to the $G < 0$ effect for high wavenumbers (modes with $\mu > 1$).

This makes [f2](#) a more discriminating test than [f1](#) for assessing scheme behaviour near non-smooth data.

7.7 Algorithmic Optimisation and Possible Extensions

The current implementation is deliberately simple and close to the mathematical schemes, but several practical improvements are natural if the code were extended to larger grids or repeated experiments.

Pre-computation of tridiagonal factors. For BTCS and CN, the left-hand matrix is constant when Δx , Δt , and α are fixed. Therefore the Thomas forward-sweep coefficients can be computed once and reused at every time step. This is done for BTCS in

`schemes/implicit.py` and for CN in `schemes/crank_nicolson.py`. This reduces the repeated linear-solve overhead while preserving exactly the same numerical method.

Memory-efficient storage. The plotting mode stores the complete history $U^0, \dots, U^{N_t}$, which costs $O(N_x N_t)$ memory. This is convenient for small teaching examples, but not necessary for production runs. A more scalable version would store only the current vector and a prescribed list of snapshot times, reducing the memory cost to $O(N_x)$ plus the selected output frames.

The practical effect was measured with `benchmark_optimizations.py` using $N_x = 1200$, $N_t = 800$, $T = 0.001$, $\alpha = 1$, `f2`, and median timings over five runs for the Thomas comparison. The baseline CN implementation rebuilds the Thomas coefficients at every time step; the optimised implementation pre-computes them once.

Table 8. Practical effect of implementation-level optimisations.

Optimisation test	Baseline	Optimised / alternative	Effect
CN Thomas solve time	≈ 0.50 s	≈ 0.30 s	$\approx 1.6\times$ faster
Max difference in final CN solution	—	0.000e+00	Same numerical result
Runtime without vs with full history	≈ 0.31 s	≈ 0.31 s	No meaningful timing penalty at this size
Stored solution memory	7.33 MiB full history	0.05 MiB for 5 snapshots	$133.5\times$ less memory

The timing gain comes from eliminating one repeated forward-sweep coefficient pass per time step. The memory result is more important for large simulations: storing the full history is acceptable for plots in this report, but selected snapshots are the better design for large N_x or N_t .

Automatic time-step control for FTCS. Since the exact explicit stability condition is $4r + \alpha \Delta t \leq 2$, the program can automatically reject an unstable explicit configuration or reduce Δt before running. This would make the solver safer for parameter sweeps and prevent wasting computation on runs that are guaranteed to diverge.

Rannacher smoothing for non-smooth data. Crank-Nicolson is unconditionally stable but can exhibit stable sign-changing oscillations for large Δt , especially near the kink of `f2`. A common remedy is to start with one or two BTCS half-steps before switching to CN. This Rannacher smoothing damps the highest-frequency components introduced by non-smooth initial data while retaining second-order behaviour for later smooth evolution.

Sparse and higher-dimensional extensions. In one dimension, the Thomas algorithm is optimal for tridiagonal systems. In two or three dimensions, the discrete Laplacian becomes sparse but no longer tridiagonal, so sparse direct solvers, conjugate-gradient-type methods, or operator-splitting approaches become more appropriate. The same operator formulation in §2.3 provides a clean path to these extensions.

Automated convergence testing. The convergence tables in §6 could be generated by a separate validation script that runs grids of (N_x, N_t) values and writes the tables directly. This would reduce manual transcription errors and make the report fully reproducible.

8. Conclusion

We implemented and validated three finite-difference schemes for the reaction-diffusion equation $u_t = u_{xx} - \alpha u$ on the unit interval. The key findings are:

1. **Stability is the dominant constraint for FTCS:** the exact condition $4r + \alpha\Delta t \leq 2$ limits practical use to coarse spatial grids or very short time intervals, unless accuracy requires small Δt independently.
2. **BTCS and CN are unconditionally stable and practically interchangeable in stability terms,** but CN achieves second-order accuracy in time at the cost of one extra $O(N_x)$ pass per step — a highly favourable trade-off.
3. **Numerical experiments confirm the main theoretical orders:** spatial convergence is approximately second-order for all schemes, while the temporal tests show first-order behaviour for BTCS and second-order behaviour for CN.
4. **The Thomas algorithm** solves the tridiagonal systems in $O(N_x)$ time with no pivoting required, making the per-step cost of BTCS and CN comparable to FTCS.
5. **The reaction term $\alpha > 0$ is benign for the implicit systems:** it improves diagonal dominance and accelerates amplitude decay. For FTCS it must still be included in the exact stability bound $4r + \alpha\Delta t \leq 2$.

For practical application, **Crank-Nicolson is recommended** as the default scheme: it combines unconditional stability, second-order accuracy, and efficient implementation via the Thomas algorithm, making it the best balance of robustness and efficiency among the three methods considered.

9. Assignment Coverage Checklist

Table 9. Coverage of the assignment questions.

Assignment item	Where it is answered
1(a) Analytical solution for f_1	§1.2
1(b) Fourier solution for f_2	§1.3
2(a)-2(d) Explicit scheme, stability, matrix form, implementation	§3.1, §4.1, §5, <code>schemes/explicit.py</code>
3(a)-3(d) Implicit scheme, stability, matrix form, implementation	§3.2, §4.2, §5, <code>schemes/implicit.py</code>
4(a)-4(e) Crank-Nicolson scheme, order, stability, matrix form, implementation	§3.3, §4.3, §5, <code>schemes/crank_nicolson.py</code>
5 Results and analysis for both initial conditions	§6.1-§6.6 and Figures 1-4

Appendix: Usage

```
# Stable comparisons for both initial conditions
python3 main.py --compare --Nx 50 --Nt 3000 --alpha 1.0 --T 0.5 --initial f1
python3 main.py --compare --Nx 50 --Nt 3000 --alpha 1.0 --T 0.5 --initial f2

# Crank-Nicolson validation figures
python3 main.py --scheme crank-nicolson --Nx 100 --Nt 500 --alpha 1.0 --T 0.5 --initial f1 --validate
python3 main.py --scheme crank-nicolson --Nx 100 --Nt 500 --alpha 1.0 --T 0.5 --initial f2 --validate

# Deliberately unstable FTCS demonstration
python3 main.py --compare --Nx 100 --Nt 500 --alpha 1.0 --T 0.5 --initial f1 --save-prefix output_unstable

# Stable explicit at Nx=100 requires Nt >= 10202 for alpha=1, T=0.5
python3 main.py --compare --Nx 100 --Nt 10202 --alpha 1.0 --T 0.5 --initial f1

# Reproduce optimisation timing and memory comparison
python3 benchmark_optimizations.py
```